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Evaluation of Microprocessors and Case Study of the Intel 8086 Microprocessor

Introduction:

Microprocessors have been the cornerstone of modern computing technology, enabling the development of everything from simple calculators to complex supercomputers. In this assignment, we will evaluate microprocessors, explore their basic parts and their functions, and conduct a case study of the widely recognized Intel 8086 microprocessor.

Q1. Evaluation of Microprocessors:

Microprocessors have been evaluated on several parameters, including performance, architecture, power consumption, and versatility:

Performance:

- **Clock Speed:** Measured in megahertz (MHz) or gigahertz (GHz), higher clock speeds generally indicate better performance.
- **Data Width:** Refers to the number of bits processed in one cycle, influencing the speed and capability of the processor.
- **Instruction Set:** The complexity and variety of instructions that the microprocessor can execute directly affect the efficiency and range of applications.

Architecture Features:

- **CISC vs. RISC:** Complex Instruction Set Computing (CISC) processors have a larger set of instructions, while Reduced Instruction Set Computing (RISC) processors focus on a smaller set for more efficient execution.
- **Multi-core Architecture:** Recent advancements allow multiple processing cores in one chip, enabling parallel processing and improved multitasking capabilities.
- **Power Consumption:**

Power efficiency is crucial for portable devices; microprocessors are often evaluated based on their power use versus performance output, leading to the development of low-power architectures.

Evolution Timeline of Microprocessors:

1. **First Generation (1971-1972) - 4-bit Microprocessors:**
   - Example: Intel 4004.
   - Featured simple instruction sets and limited processing power.
2. **Second Generation (1973-1978) - 8-bit Microprocessors:**
   - Example: Intel 8080, Motorola 6800.
   - Improved speed and instruction sets, introduced external memory and I/O support.
3. **Third Generation (1978-1985) - 16-bit Microprocessors:**
   - Example: Intel 8086, Motorola 68000
   - Enhanced computing power, larger memory addressing, and support for multitasking.
4. **Fourth Generation (1985-1990) - 32-bit Microprocessors:**
   - Example: Intel 80386, Motorola 68020.
   - Enabled advanced computing with features like pipelining and memory management.
5. **Fifth Generation (1990-Present) - 64-bit Microprocessors:**
   - Example: Intel Core series, AMD Ryzen series.
   - Enhanced parallel processing, multi-core architectures, and higher clock speeds.

Microprocessor	Year	Word Length	Memory	Pins	Clock
4004	1971	4-bit	1 KB	16	750kHz
8085	1976	8-bit	64 KB	40	3-6 MHz
8086	1978	16-bit	1MB	40	5-8 MHz
80286	1982	16-bit	16MB real 4 GB virtual	68	6-12.5 MHz
80386	1985	32-bit	4GB real 64TB virtual	132 14X14 PGA	20-33 MHz
80486	1989	32-bit	4GB real 64TB virtual	168 17X17 PGA	25-100 MHz

Pentium	1993	32-bit	4GB real 32-bit address 64-bit data bus	237 PGA	60-200
Pentium Pro	1995	32-bit	64GB real 36-bit address bus	387 PGA	150-200 MHz
Pentium II	1997	32-bit	–	–	233-400 MHz
Pentium III	1999	32-bit	64GB	370 PGA	600-1.3 MHz
Pentium 4	2000	32-bit	64GB	423 PGA	600-1.3 GHz
Itanium	2001	64-bit	64 address lines	423 PGA	733 MHz-1.3 GHz

Q2. Basic Parts of Microprocessor and Their Working:

A microprocessor is composed of several key components, each playing a crucial role in its overall function:

2.1 Architecture:

Divided into two functional units:

1. **BIU (Bus Interface Unit):** Handles external communications, instruction fetching, and queue management. The BIU is responsible for fetching instructions from memory, decoding them, and managing data transfer between the microprocessor and memory or I/O devices. It also maintains a 6-byte pre-fetch instruction queue to support pipelining.
2. **EU (Execution Unit**)**:** . The EU executes the instructions fetched by the BIU, performs arithmetic and logical operations using the Arithmetic Logic Unit (ALU), and manages the general-purpose and special-purpose registers[1].

Arithmetic Logic Unit (ALU):

- **Function:** The ALU performs arithmetic operations such as addition, subtraction, multiplication, division, and logical comparisons (AND, OR, NOT).
- **Working:** Data is fed into the ALU, and based on the operation specified by the instruction set, it processes this data and outputs the result.

2.2 Control Unit (CU):

Function: The CU orchestrates the operations of the microprocessor by directing the flow of data between the ALU, registers, and memory.

Working:

- It fetches instructions from memory, Decodes them.
- Coordinates operations of all processor components.
- Manages timing and execution sequence.
- Directs data flow between processor components.

2.3 Registers:

Small, high-speed storage locations within the CPU .

Types include:

- **General Purpose Registers (AX, BX, CX, DX):** Used for various data manipulation tasks.
- **Segment Registers (CS, DS, SS, ES):** Divide memory into segments, aiding in memory management.
- **Index Registers (SI, DI):** Used for indexed addressing and string operations.
- **Pointer Registers (BP, SP, IP):** Base Pointer (BP) and Stack Pointer (SP) help with stack operations; Instruction Pointer (IP) holds the address of the next instruction to be executed.

Working: They provide quick access to data that the CPU processes, speeding up computation by minimizing the need to access slower main memory.

Memory Segmentation:

- Used segment:offset addressing.
- Memory divided into 64KB seg.ments
- Physical address = Segment address × 16 + Offset

2.4 Bus Interface Unit:

Function: The bus interface facilitates communication among the CPU, memory, and peripheral devices.

Working:

- Manages communication between the CPU and external components.
- Handles data, address, and control buses.
- Synchronizes data transfer.

2.5 Cache Memory:

Function: Cache memory is a small-sized type of volatile memory that provides high-speed data access to the processor.

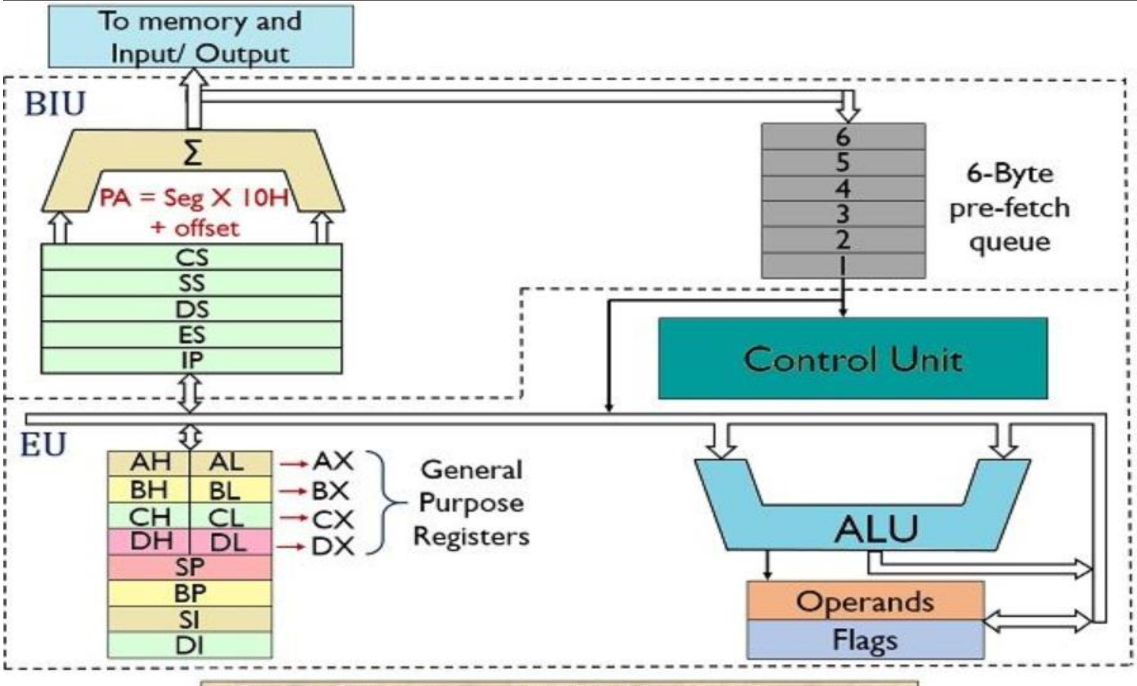
Working: By storing frequently accessed data and instructions, the cache reduces the average time to access data from the main memory, thereby enhancing performance.

2.6 Flags Register:

- Contains status flags that indicate the state of the processor and the outcomes of operations (e.g., zero flag, carry flag, overflow flag).

2.7 Clock:

- Provides timing signals to synchronize operations.
- Determines processor speed.



Block Diagram of 8086 Microprocessor

Q3. Case Study of the 8086 Microprocessor:

The Intel 8086 microprocessor, released in 1978, marked a significant advancement in computing technology, being one of the first widely used 16-bit microprocessors.

3.1 Overview:

- Introduced by Intel in 1978.
- 16-bit processor (16-bit data bus).

- 20-bit address bus (could address up to 1MB of memory).
- Operating frequency: 5-10 MHz.
- ~29,000 transistors.
- First x86 architecture processor with commercial success.

16-bit Microprocessor: The 8086 can process 16 bits of data simultaneously, allowing for larger data handling than its 8-bit predecessors.

Segmentation: It introduced memory segmentation, allowing it to address up to 1 MB of memory by dividing the memory into segments for code, data, and stack.

3.2 Architecture:

Divided into two functional units:

- **BIU (Bus Interface Unit):** Handles external communications, instruction fetching, and queue management.
- **EU (Execution Unit):** Executes instructions, performs calculations.

Registers:

1) **General Purpose Registers:** AX, BX, CX, DX (each 16 bits, can be accessed as 8-bit registers: AH/AL, BH/BL, etc.)
2) **Pointer and Index Registers:** SP (Stack Pointer), BP (Base Pointer), SI (Source Index), DI (Destination Index).
3) **Segment Registers:** CS (Code), DS (Data), SS (Stack), ES (Extra).
4) **Instruction Pointer (IP):** Points to next instruction.
5) **Flag Register:** Contains status flags (carry, zero, sign, etc.)

Case Study Example:

In a practical application, the 8086 microprocessor was used in the IBM PC, which revolutionized personal computing. Its architecture allowed for the development of MS-DOS, an operating system that became the foundation for many future software applications. The combination of the 8086's performance and the flexibility of MS-DOS made the IBM PC a commercial success and established Intel's dominance in the microprocessor market.

3.3 Instruction Set:

- Over 20,000 instruction variations.

- Supported various addressing modes.
- Included data transfer, arithmetic, logical, string, and control transfer instructions.

3.4 Legacy:

- Established the foundation for the x86 architecture.
- Compatibility with 8086 instruction set remains in modern processors.
- Influenced subsequent processor designs (80186, 80286, 80386, etc.)

3.5 Operating Modes:

Minimum Mode: Used in simple systems, with a single processor managing all functions.

Maximum Mode: Designed for multiprocessor configurations, allowing multiple CPUs to operate seamlessly.

3.6 Applications:

The 8086 was widely employed in early personal computers, including the IBM PC and compatible systems, which helped to define the standard for PC architecture.

3.7 Conclusion:

The 8086 microprocessor was a pivotal innovation in computer hardware, setting the foundation for subsequent processor architectures. Its 16-bit architecture, memory segmentation, and versatile instruction set allowed it to thrive in a range of applications, making it a landmark in the evolution of microprocessors.

Conclusion:

Microprocessors have transformed the landscape of computing through significant advancements in performance, architecture, and efficiency. Understanding their basic parts and evaluating specific case studies like the Intel 8086 provides valuable insights into the continued evolution of technology and the foundation for modern computing. As we look forward, the principles established by early microprocessors still inform contemporary designs, which will lead to even greater capabilities and applications in the future.